Tamanjyot Singh

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Bio: **Aerospace engineer** by training, pursuing a **Doctorate in Physics** with specialization in astrophysics. Interested in understanding black hole spin dynamics and their implications. Currently working as a research assistant at the University of Texas at Dallas under the supervision of Dr. Michael Kesden. My thesis work focuses on (i) **inclination distributions in tidal disruption events by spinning supermassive black holes** and their observational implications, (ii) **precessional and nutational effects on the gravitational wave signals** originating from inspiralling stellar mass binary black hole as currently observed by LIGO, and (iii) **degeneracies between lensed and precessing gravitational wave signals** for current LIGO sensitivities.

Education

The University of Texas at Dallas — Richardson, TX, USA
Doctorate in Physics — *Expected May 2026*

Thesis: The Black Hole Ballet: Spin and precession effects on gravitational waves and tidal disruption Events

Advisor: Dr. Michael Kesden

The University of Texas at Dallas — Richardson, TX, USA
Master of Science in Physics, GPA: 3.93/4.0 — *December 2023*

Punjab Engineering College (Deemed to be University) — Chandigarh, India
Bachelor of Technology in Aerospace Engineering with Minor Specialization in Mathematics, GPA : 9.2/10.0, Rank 2 — *June 2020*

Major project: Computational study of flame propagation in a micro-scale channel and applications of such micro-combustors in thermo-electric power generation

Experience

Research Assistant — May 2022 - Present
The University of Texas at Dallas — *TX, USA*

Supervisor: Dr. Michael Kesden

- **Lensing and Precession**: Distinguishing between a lensed and precessing gravitational wave signal, exploring degeneracies and establishing differences.
  Meeting weekly, developing a code, mentoring a post-baccalaureate student, and preparing a manuscript to be submitted for publication.
- **Precession in gravitational waves**: Investigating the effects of spin dynamics in a binary black hole system on the gravitational wave emission.
  Meeting weekly, developing a code, working in a team environment, mentoring undergraduate students, and preparing a manuscript to be submitted for publication.
- **Tidal disruption events**: Solving for inclination biases induced by spin breaking the spherical symmetry in stellar disruptions by supermassive black holes.
  Meeting weekly, developing a code, mentoring undergraduate students, preparing, and submitting a manuscript resulting in a published research article in Physical Reviews.

Teaching Assistant — January 2021 - May 2022
The University of Texas at Dallas — *TX, USA*

- **Physics Laboratory I: Mechanics**: Hosting weekly sessions for students to go through the Mechanics Lab experiments. Grading their Lab report submissions

Research Intern — January 2019 - April 2019
Combustion research Laboratory, IIT Bombay — *Mumbai, India*

- **Combustion characteristics and the effect of hydrogen content on syngas-air mixtures**: Formulated an Analytical correlation for Laminar Burning Velocity of Multi-component Gaseous Mixtures Simultaneously at Elevated Pressures and Temperatures.

IAF (Indian Air Force) Intern — June 2018
3 BRD, Air Force Station — *Chandigarh, India*

Publications

Published papers:

Tamanjyot Singh and M. Kesden (2024, February): "Distribution of orbital inclinations for tidal disruption events by Kerr black holes." 109 (4), 043016

Talks

- Fall meeting 2024, Texas section APS, Southern Methodist University, Dallas, TX — October 2024
  Contributed Talk: Signatures of generic precession in gravitational wave signals from merging binary black holes
- Center for Computational Astrophysics, Flatiron Institute, New York City, NY — May 2024
  Invited visit: Precession in GWs
- APS April meeting 2024, Sacramento, CA — April 2024
  Contributed Talk: Gravitational wave signals from precessing binary black hole inspirals
- TEXAS PAARE Annual Symposium (UTD/UNT symposium) 2024, UNT, Denton, TX — March 2024
  Contributed Talk: Tidal disruption events: Inclination distributions for spinning supermassive black holes
- 10th Gulf Coast Gravity meeting 2024, UT Austin, TX — March 2024
  Contributed Talk: Gravitational wave signals from precessing binary black hole inspirals
- International Conference on Gravitation and Cosmology 2023, IIT Guwahati, Assam, India — December 2023
  Contributed Talk: Inclination distributions in tidal disruption events by spinning supermassive black holes
- Fall meeting 2023, Texas section APS, Angelo State University, San Angelo, TX — October 2023
  Contributed Talk: Identifying Precession in Gravitational Wave signals from Binary Black Hole inspirals
- Spring meeting 2023, Texas section APS, Texas A&M University, Commerce, TX — March 2023
  Contributed Talk: Tidal disruption events by spinning supermassive black holes

Notable Achievements

- **APS April meeting DGRAV Travel Award** 2024
- **Margie Renfrow travel award** 2024 by the Department of Physics, The University of Texas at Dallas, TX
- **Teaching Assistant award** 2022 at The University of Texas at Dallas, TX
- ***PECOSA Sydney Prize*** for year 2017-18 (Merit Based - Awarded to 3 out of 700+ students)
- Secured **1st position** in the Mathematics Quiz sponsored by Punjab State Council for Science and Technology, Chandigarh (Gold Medal - District Level Competition)
- **99.3%ile** in JEE (Mains) 2016 taken by 1,400,000+ candidates all over India
- **International Rank 76** in IOS (International Olympiad of Science) 2013 (Gold Medal + *Special Achievement Certificate*)

Workshops and Extra Course Certificates

- Attended **Entering Mentoring** at the University of Texas at Dallas (July 2024)
- Course Certificate for **Astro 101 : Black Holes** by University of Alberta on Coursera. (June 2020)
- Hosted a workshop at undergrad level for **Gravitational lensing** at Punjab Engineering College, Chandigarh (October 2019)
- Attended **ASTROWIN** , Winter School on Astronomy at the Birla Museum, Hyderabad (February 2019)
- Course Certificate for **Particle Physics: an Introduction** by University of Geneva on Coursera. (July 2018)
- Course Certificate for **Understanding Einstein: The Special Theory of Relativity** by Stanford University on Coursera. (July 2017)

Skills

- **Programming Languages**: Python, C++, SQL, Matlab
- **Softwares**: pyLIMA, Fluent, AutoCAD, OriginLab, Athena++, Jupyter, Solidworks and ANSYS

Responsibilities

- **GSP (Graduate Students in Physics) Astronomy committee head**: Grad student liaison for all things telescope, hosting star-watch and lunar eclipse parties. Responsible for maintenance of the existing and buying new equipment. (March 2022 - February 2024)
- **Vice President GSP**: Responsibilities included shouldering the leadership workload of the president and respresenting the graduate student body concerns to the physics department. (May 2023 - July 2024)
- **Journal Club host**: Hosting weekly meetings with students across DFW in a joint astronomy/astrophysics journal club with University of North Texas, Denton. Graduate students encouraged to present a recent paper in their field of research (of their choice) at least once per semester. (August 2024 - January 2025)
- **Course Facilitator**: Volunteered to teach and translate gurbani (Sikh holy text, written in Punjabi) along hosting with biweekly community sessions, fostering an environment of positive outlook on life as Sikhs in USA. (August 2024 - May 2025)

Mentoring Experience

- **Allison Powell**: Summer REU student 2022
- **Niel Shah**: Summer REU student 2023
- **Deven Linthicum**: Summer REU student 2023
- **Tien Nguyen**: PAARE fellow 2023-24: post-baccalaureate student
- **Maya Basu**: Summer REU student 2024
- **Tinatin Tsiskaridze**: Summer exchange student 2024 from Tblisi State University, Georgia